SDS8102: Advanced Mathematical Statistics

Week 12 – Lecture 1
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Lecturer: Alexandre Tsybakov

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1 Outline

We will study the following two problems.

1. Density estimation.

   We have a sample $X_1, ..., X_n$ of i.i.d. random variables distributed with density $p(\cdot)$ w.r.t. the Lebesgue measure in $\mathbb{R}^d$.

   Problem: estimate p from these data.

2. Regression.

   We have a sample $(X_i, Y_i), i = 1, ..., n$ of independent pairs of random variables (in this course we mainly assume that X_i's are deterministic). Here, $X_i \in \mathcal{X}$, $Y_i \in \mathbb{R}$, and $\mathcal{X}$ is a general space. Assume that $Y_i = f(X_i) + \xi_i$, where f is an unknown function from $\mathcal{X}$ to $\mathbb{R}$, and ξ_i's are zero mean random variables.

   Problem: estimate f.

The nonparametric nature of these problems means that we do not have much knowledge about p, f and the distribution of (X, Y).

2 The problem of density estimation

Let $X_1, ..., X_n$ be i.i.d. real valued random variables with density p w.r.t. the Lebesgue measure on $\mathbb{R}$. The considered problem is to estimate p given nonparametric prior information about p. In the classical parametric approach, we know that the density is from a parametric family: $p(x) = g(x, \theta)$, where g is a known function, and $\theta \in \mathbb{R}^m$ is unknown. Thus the estimation of p is equivalent to the estimation of θ. If $\hat{\theta}_n$ is an estimator of θ, then $g(x, \hat{\theta}_n)$ is an estimator of p.

In nonparametric approach, we only assume that we have a prior information of the form: $p \in \mathcal{P}$, where $\mathcal{P}$ gives a wide (or nonparametric) class of densities.

Examples:

- $\mathcal{P}$ ={all continuous densities}.
- $\mathcal{P}$ ={all densities such that $|p(x) - p(x')| \le L|x - x'|, \forall x, x'$}.
- $\mathcal{P}$ = {all densities that are monotonically decreasing on $[0, \infty)$}.

How to estimate p nonparametrically? A very basic solution is the histogram. Assume for simplicity that $p : [0,1] \to \mathbb{R}$. Divide the interval $[0,1]$ into segments $\{A_j\}_{j=1}^m$, called the bins of histogram, with equal length $h = 1/m$. Define n_j to be the number of data points that fall into A_j:

$$n_j = \sum_{i=1}^{n} I(X_i \in A_j).$$

After normalization, the histogram density estimator is defined as

$$\begin{aligned}\hat{p}_n^H(x) &= \sum_{j=1}^{m} \frac{n_j}{nh} I(x \in A_j) \\ &= \frac{1}{nh} \sum_{i=1}^{n} \sum_{j=1}^{m} I(X_i \in A_j, x \in A_j) \\ &=: \frac{1}{nh} \sum_{i=1}^{n} K_h(X_i, x).\end{aligned}$$

Exercise: Show that $\hat{p}_n^H$ is a density itself for any fixed $X_1, ..., X_n$, i.e.,

$$\hat{p}_n^H \geq 0, \ \forall x \text{ and } \int \hat{p}_n^H(x)dx = 1.$$

Remarks:

- The estimator depends on one parameter $h > 0$ (smoothing parameter).
- What happens if we change h?

  When h is small, the estimator will take the same values around the data points (under-smoothing).

  When h is large, the entire interval becomes one bin. The estimator is a constant (over-smoothing).

  The question of optimal estimation is the question of choosing the optimal h between the two extremes.
- The histogram density estimator is piecewise constant and discontinuous. Therefore it is not an ideal estimator for a continuous density.

3 Kernel density estimator

Suppose $F(x)$ is the c.d.f. of X_i's, then

$$p(x) = F'(x) \text{ a.e.}$$

An estimator for $F(x)$ is the empirical distribution function

$$F_n(x) = \frac{1}{n} \sum_{i=1}^{n} I(X_i \leq x)$$

that is unbiased,

$$\mathbb{E}F_n(x) = \mathbb{P}(X_i \leq x) = F(x),$$

consistent (by the law of large numbers):

$$\forall x, \qquad F_n(x) \xrightarrow{a.s.} F(x), \text{ as } n \to \infty,$$

and uniformly consistent:

$$\sup_{x \in \mathbb{R}} |F_n(x) - F(x)| \xrightarrow{a.s.} 0, \text{ as } n \to \infty \text{ (Glivenko-Cantelli theorem).}$$

Despite all these nice properties, $F_n(x)$ is only an estimator for $F(x)$ while our goal is to estimate the density $p(x)$. Rosenblatt(1956) came up with the following estimator:

$$\hat{p}_n^R(x) = \frac{F_n(x + h/2) - F_n(x - h/2)}{h}.$$

Intuitively, $h > 0$ has to be small for this approximation to make sense. The Rosenblatt estimator could be further written as

$$\frac{1}{nh} \sum_{i=1}^{n} I(x - h/2 < X_i \le x + h/2).$$

If we write

$$K_0(u) = I(-1/2 < u \le 1/2)$$

then

$$\hat{p}_n^R(x) = \frac{1}{nh} \sum_{i=1}^{n} K_0\Big(\frac{X_i - x}{h}\Big).$$

Notice that like the histogram estimator, the Rosenblatt estimator is still not smooth. A generalization of the Rosenblatt estimator is due to Parzen(1962):

$$\hat{p}_n(x) = \frac{1}{nh} \sum_{i=1}^{n} K\Big(\frac{X_i - x}{h}\Big)$$

where $K : \mathbb{R} \to \mathbb{R}$ is integrable and (typically) satisfies $\int K(u)du = 1$. Note that K is not necessarily nonnegative. The function K is called a kernel, and h is called a bandwidth.

Examples:

- $K(u) = \frac{1}{2} I(|u| \le 1)$ (the rectangular kernel)
- $K(u) = \frac{3}{4}(1 - u^2)_+$, where $x_+ = \max(0, x)$ (the parabolic kernel, or the Epanechnikov kernel)
- $K(u) = \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}}$ (the Gaussian kernel)
- $K(u) = \frac{\sin u}{\pi u}$ (the sinc kernel)

Remarks:

- All probability densities are kernels. But the reverse is not true. For example, the sinc kernel is not integrable, but we will see that it has very good properties.
- If K is integrable with $\int K(u)du = 1$ and K is nonnegative. Then $\int \hat{p}_n(x)dx = 1$ and $\hat{p}_n(x) \ge 0$, which means the estimator itself is a density, for all $X_1, ..., X_n$ fixed.

- The estimator $\hat{p}_n(x) = \frac{1}{nh}\sum_{i=1}^n K\Big(\frac{X_i - x}{h}\Big)$ is a sum of "bells". A "bell" is a function of the form

$$x \mapsto \frac{1}{nh} K\Big(\frac{X_i - x}{h}\Big).$$

  When h is small, the estimator consists of separate and identical "bells" at each data point; when h is large, the estimator is small everywhere and flat. Both situations are not satisfying. The optimal h is somewhere between these two extremes. Our aim now will be to quantify these effects in mathematical terms and characreize the optimal choice of h.

4 Statistical properties of kernel estimators

First, we want to control the estimation error at a fixed point x: $\hat{p}_n(x) - p(x)$. We introduce a non-random characteristic (mean squared error) at point x:

$$MSE = \mathbb{E}((\hat{p}_n(x) - p(x))^2)$$

where the expected value is taken w.r.t. $X_1, ..., X_n$. To be specific,

$$\begin{aligned}
&\hat{p}_n(x) = \hat{p}_n(x, X_1, ..., X_n),\\
&MSE = \int ... \int (\hat{p}_n(x, X_1, ..., X_n) - p(x))^2 \prod_{i=1}^n p(X_i) dX_i.
\end{aligned}$$

Remarks:

- Note that x is fixed in the definition of MSE.
- There is a bias-variance decomposition. If $\mathbb{E}\hat{p}_n^2(x) < \infty$, we can always write

$$MSE = b^2(x) + \sigma^2(x)$$

  where the bias $b(x) = \mathbb{E}\hat{p}_n(x) - p(x)$, and the variance

$$\sigma^2(x) = \mathbb{E}((\hat{p}_n(x) - \mathbb{E}\hat{p}_n(x))^2).$$

  Indeed, let ξ be a random variable such that $\mathbb{E}\xi^2 < \infty$, $c \in \mathbb{R}$ is a constant. Then it is easy to show that

$$\mathbb{E}((\xi - c)^2) = (\mathbb{E}\xi - c)^2 + \mathbb{E}((\xi - \mathbb{E}\xi)^2)$$

  Taking here $\xi = \hat{p}_n(x)$, $c = p(x)$ and obtain the desired decomposition.

We will study separately $b(x)$ and $\sigma^2(x)$.

Proposition 1. *Assume that the density is bounded by a constant, $p(x) \le p_{max}$, $\forall x \in \mathbb{R}$, and $\int K^2(u)du < \infty$. Then we have for all $x \in \mathbb{R}$, $n \ge 1$, $h > 0$,*

$$\sigma^2(x) \le \frac{C}{nh}$$

where $C = p_{max} \int K^2(u)du$.

Remarks:

- The bound on the variance is a non-asymptotic bound that holds for all n. It is also a uniform bound on the real line that does not depend on the choice of x.
- If we want $\sigma^2(x)$ to be small, we need nh to be large. In fact, in order to make $\sigma^2(x)$ tend to 0 as n goes to infinity, h should have order higher than $1/n$.

Proof: Write

$$\hat{p}_n(x) - \mathbb{E}\hat{p}_n(x) = \frac{1}{nh}\sum_{i=1}^{n}\eta_i(x)$$

where the random variables η_i's are defined as

$$\eta_i(x) = K\left(\frac{X_i - x}{h}\right) - \mathbb{E}K\left(\frac{X_i - x}{h}\right).$$

The η_i's are i.i.d. because the X_i are i.i.d. Furthermore, $\mathbb{E}\eta_i(x) = 0$, and

$$\begin{aligned}\mathbb{E}\eta_i^2(x) \leq& \mathbb{E}K^2\left(\frac{X_i - x}{h}\right)\\ =& \int K^2\left(\frac{z-x}{h}\right)p(z)dz\\ \leq& p_{max}h\int K^2(u)du.\end{aligned}$$

Now we can bound the variance:

$$\begin{aligned}\sigma^2(x) =& \mathbb{E}((\frac{1}{nh}\sum_{i=1}^{n}\eta_i(x))^2)\\ =& \frac{1}{n^2h^2}\sum_{i=1}^{n}\mathbb{E}\eta_i(x)^2\\ \leq& \frac{1}{nh}p_{max}\int K^2(u)du.\end{aligned}$$

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When studying bias, we need more assumptions.

Definition. (Hölder class of functions) *Let $\beta > 0$, $L > 0$, let $l = \lfloor\beta\rfloor$ (Here the floor function is defined as the maximal integer that is strictly less than β). The Hölder class $\Sigma(\beta, L)$ is the class of all functions $f : \mathbb{R} \to \mathbb{R}$, s.t. $f^{(l)}$ exists and*

$$|f^{(l)}(x) - f^{(l)}(x')| \leq L|x - x'|^{\beta - l},\ \forall x, x' \in \mathbb{R}.$$

Examples:

- Take $\beta = 1$, then $l = 0$. The Hölder condition takes the form

  $$|f(x) - f(x')| \leq L|x - x'|,\ \forall x, x' \in \mathbb{R}.$$

  Thus the Hölder class $\Sigma(1, L)$ is composed of all Lipschitz functions with Lipschitz constant less than or equal to L.

- Take $\beta = 2$, then $l = 1$, The Hölder condition is

$$|f'(x) - f'(x')| \leq L|x - x'|.$$

With the definition of the Hölder class, we can now specify the class of densities that p belongs to. Define

$$\mathcal{P}(\beta, L) = \{p \in \Sigma(\beta, L) : p \geq 0, \int p = 1\}.$$

Definition. *A function $K : \mathbb{R} \to \mathbb{R}$ is called a kernel of order l ($l \geq 1$ is an integer) if the functions $u \mapsto u^j K(u), j = 0, ..., l$ are integrable and*

$$\int K(u)du = 1, \text{ and } \int u^j K(u)du = 0, j = 1, ..., l.$$

Example: If $l = 1$ then a kernel of order l is any function that satisfies

$$\int K(u)du = 1 \text{ and } \int uK(u)du = 0.$$

This is true, for instance, if K is a symmetric density.

Proposition 2. *Let $p \in \mathcal{P}(\beta, L)$ with $\beta > 0$, $L > 0$. Let K be of order $l = \lfloor \beta \rfloor$ and such that $\int |u|^\beta |K(u)|du < \infty$. Then for all $x \in \mathbb{R}$, $n \geq 1$, $h > 0$, we have*

$$|b(x)| \leq C_1 h^\beta$$

where $C_1 = \frac{L}{l!} \int |u|^\beta |K(u)|du$.

Before proving the proposition, notice that for this upper bound to tend to zero as n goes to infinity, h needs to go to zero. But since we also need to control the variance, h cannot go to zero arbitrarily fast.

Proof: We can write the bias in the form

$$b(x) = \frac{1}{h} \int K\left(\frac{z - x}{h}\right) p(z)dz - p(x).$$

With a change of variable, the expression is also equal to

$$\int K(u)(p(x + uh) - p(x))du.$$

Because K is a kernel of order l, there are two conditions we can use:

$$\int K(u)du = 1 \text{ and } \int u^j K(u)du = 0, j = 1, ..., l.$$

Since $p \in \mathcal{P}(\beta, L)$, the lth derivative of p exists, and

$$p(x + uh) = p(x) + p'(x)uh + .. + \frac{(uh)^l}{l!} p^{(l)}(x + \tau uh), \quad 0 \leq \tau \leq 1.$$

Plug this back into the expression of b, and the terms with order strictly less than l will integrate to 0 because K is of order l, leaving

$$\begin{aligned} b(x) &= \int K(u)\frac{(uh)^l}{l!}p^{(l)}(x+\tau uh)du \\ &= \int K(u)\frac{(uh)^l}{l!}(p^{(l)}(x+\tau uh)-p^{(l)}(x))du. \end{aligned}$$

Since we have a Lipschitz condition on $p^{(l)}(x)$,

$$|p^{(l)}(x+\tau uh)-p^{(l)}(x)| \le L|\tau uh|^{\beta-l} \le L|uh|^{\beta-l}.$$

The bias is thus bounded by

$$|b(x)| \le \frac{Lh^\beta}{l!}\int |u|^\beta |K(u)|du.$$

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Combining Propositions 1 and 2, we have a bound on the MSE:

$$MSE = b^2(x)+\sigma^2(x) \le C_1^2 h^{2\beta} + \frac{C}{nh}.$$

The right hand side is minimized at

$$h_{opt} = \Big(\frac{C}{2\beta C_1^2 n}\Big)^{\frac{1}{2\beta+1}} \sim n^{-\frac{1}{2\beta+1}}.$$

As h increases, the bias goes up and the variance goes down. So the choice of h is a result of the bias-variance trade-off.
Plugging it back into the bound on MSE, we can get $MSE_{opt} = O(n^{-\frac{2\beta}{2\beta+1}})$, which is the rate of convergence of the kernel estimator. Here is a formal statement of the theorem.

Theorem 1. *Assume that the kernel K satisfies the assumptions of Propositions 1 and 2, also assume that $h = \alpha n^{-\frac{1}{2\beta+1}}$ for some positive α. Then the kernel density estimator $\hat{p_n}$ satisfies*

$$\sup_{x\in\mathbb{R}}\sup_{p\in\mathcal{P}(\beta,L)} \mathbb{E}((\hat{p}_n(x)-p(x))^2) \le C_* n^{-\frac{2\beta}{2\beta+1}}$$

where C_* is a constant that only depends on β, L, α and the kernel $K(\cdot)$.

Proof: The only thing left to show is that there is a uniform upper bound for the family of densities in $\mathcal{P}(\beta, L)$, i.e., we need

$$p(x) \le p_{max} < \infty, \ \forall p \in \mathcal{P}(\beta,L), x \in \mathbb{R}.$$

Following the proof of Proposition 2, where we take the bandwidth $h = 1$, and take $K = K_*$ to be a bounded kernel of order l, we obtain that the bias will be

$$|\int K_*(z-x)p(z)dz - p(x)| \le C_1 h^\beta = C_1.$$

From this we obtain a uniform bound for the density

$$p(x) \le C_1 + |\int K_*(z-x)p(z)dz| \le C_1 + \max_x |K_*(x)| \int p(z)dz.$$

Since p is a density, $\int p(z)dz = 1$, this bound is uniform over all densities in $\mathcal{P}(\beta, L)$. ■